

Zhenjiang Promotional Website Design Document

1. Executive Summary

This document presents the complete design plan for the Zhenjiang City Promotional Website. With Jiangnan Ink Wash + Modern Minimalism as the core style, incorporating a warm color palette and Cormorant Garamond + Inter font combination, the website features an 8-section single-page scrolling structure. Using HTML5/CSS3/Native JS to implement scroll animations, 3D Earth, and video playback interactions, creating a city promotional platform that combines 3,000 years of historical heritage with contemporary artistic sensibility.

2. Detailed Summary

2.1 Project Overview

This project is the Zhenjiang City Promotional Website, designed to showcase Zhenjiang's 3,000-year historical heritage, Three Mountains scenery (Jinshan, Jiaoshan, Beigu), poetry culture, White Snake legend, culinary culture, and modern development to visitors through modern web design.

2.2 Design Style Positioning

(1) Overall Style

Jiangnan Ink Wash + Modern Minimalism, blending Jiangnan garden aesthetics with modern design, emphasizing white space, breathing room, and visual hierarchy to create a warm and elegant browsing experience.

(2) Color System

Color Name	Hex Code	Usage
Warm Cream	#fdfbf7	Main Background
Warm Gray-White	#f4f0ea	Alternate Background
Ink Brown	#2c241b	Primary Text
Warm Gray	#7a7065	Secondary Text

Vermillion Red	#9e3d32	Accent Color
Warm Gold	#b8860b	Decorative Color

(3) Font Selection

Usage	Font	Characteristics
Headings	Cormorant Garamond	Elegant serif, classical temperament
Body	Inter	Modern sans-serif, highly readable

2.3 Page Structure Design

(1) 8 Main Sections (Single-Page Scrolling)

1. Hero - Full-screen background image, title animation, scroll indicator
2. History - 3,000-year history introduction, statistics
3. Mountains - Jinshan, Jiaoshan, Beigu card displays
4. Poetry - Poetry quotes from Wang Anshi and Xin Qiji
5. Legends - White Snake legend cultural introduction
6. Culture - Zhenjiang fragrant vinegar, pot cover noodles, and other cuisine
7. Modern - Runyang Bridge video showcase
8. Location - 3D Earth displaying geographic location

(2) Responsive Breakpoints

Device	Width	Layout
Mobile	<768px	Single column, smaller fonts
Tablet	768-1024px	Two columns, moderate white space
Desktop	>1024px	Full layout

2.4 Interaction Design

(1) Animation Effects

- **fadeIn** - Element fade-in and slide-up animation
- **parallax** - Hero section parallax scrolling
- **hover scale** - Image hover zoom effect
- **float** - Floating icon animation

(2) 3D Earth

- Using **COBE** library to render 3D Earth
- Marking Zhenjiang's geographic location (32.20°N, 119.45°E)
- **Mouse drag** for free rotation viewing
- Auto-rotation display, pauses during interaction

(3) Video Playback

- Runyang Bridge video showcase
- Auto-play when entering viewport
- Custom play/pause button
- Muted loop playback

2.5 Technical Implementation

(1) Tech Stack

HTML5 + CSS3 + Native JavaScript, no framework dependencies, balancing lightweight and extensibility.

(2) Core Features

- **Scroll Animations** - Intersection Observer API
- **3D Earth** - COBE library + mouse event listeners
- **Video Control** - HTML5 Video API + custom buttons
- **Parallax Effect** - scroll event + requestAnimationFrame

(3) Performance Optimization

- Image lazy loading
- Font preconnection (Google Fonts)
- Animation GPU acceleration (will-change)
- Noise texture background (SVG data URI)

2.6 Design Highlights

- **Noise Texture** - Full-screen noise overlay for added texture
- **Warm Color System** - Warm and elegant color scheme
- **Cultural Integration** - White Snake legend, 3,000-year history, poetry culture
- **3D Interaction** - Interactive Earth displaying geographic location

2.7 Design Summary

With **Jiangnan Ink Wash · Modern Interpretation** as the core, combining warm colors, elegant fonts, and smooth interactions to create a city promotional website that balances cultural heritage with modern sensibility.

3. Key Questions & Answers

1. What is the core design style and color logic of this website?

Answer: The core style is **Jiangnan Ink Wash + Modern Minimalism**; the color palette with **warm cream and ink brown** creates a warm and elegant atmosphere, **vermillion red** represents historical heritage, and **warm gold** adds sophistication, forming a unified and sophisticated visual system.

2. What is the page structure and responsive rules of the website?

Answer: Adopting **single-page scrolling design** with **8 core sections**; responsive adaptation by breakpoints: **<768px single column, 768-1024px two columns, >1024px full layout**, ensuring multi-device experience.

3. How is the 3D Earth feature implemented?

Answer: Using **COBE** library to render WebGL Earth, marking Zhenjiang's coordinates, supporting **mouse drag rotation** to view from different angles, auto-rotation resumes after 3 seconds.

4. What are the tech stack and core features?

Answer: Tech stack is **HTML5 + CSS3 + Native JS**, no framework dependencies; core features include **scroll animations, 3D Earth interaction, and video playback control**.

4. File Structure

```
app/
├── index.html           # Main page
├── styles.css           # Stylesheet (custom CSS)
├── script.js           # Interaction scripts
├── design.md           # Design document
├── images/             # Image resources
│   ├── hero.jpeg
│   ├── 1.jpg
│   ├── 2.jpg
│   └── f1.jpg
```

```
├── f2.jpg
├── f3.jpg
├── f4.jpg
└── video/                                # Video resources
    ├── cover-1775886466299.jpg
    └── 01e951292921cebd4f0370019b65099f83_258.mp4
```

5. Summary

Project Achievements

The Zhenjiang Promotional Website successfully integrates **3,000 years of history and culture** with **modern web technology**, achieving design goals through the following core elements:

Dimension	Implementation
Visual Design	Warm color palette creates Jiangnan ink wash aesthetic, noise texture adds texture
Interactive Experience	Smooth interactions including 3D Earth drag rotation, parallax scrolling, video playback control
Cultural Expression	Deep integration of Three Mountains scenery, White Snake legend, poetry culture
Technical Implementation	Native tech stack ensures lightweight efficiency, no framework dependencies

Design Value

- Cultural Heritage** - Interpreting Zhenjiang's historical legacy through modern design language
- User Experience** - Single-page scrolling architecture provides immersive browsing experience
- Technical Elegance** - Rich interactive effects achieved with concise code
- Brand Building** - Unified visual system strengthens city image

Future Outlook

- Add multi-language support (English, Japanese, etc.)
- Integrate map navigation features
- Add visitor comment and interaction modules

- Optimize mobile touch experience